

PhaseGAN: High-Fidelity Vocoder via Decoupled Amplitude and GAN-Driven Phase Reconstruction

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Abstract. A vocoder is a pivotal component of modern text-to-speech (TTS) systems. Despite the significant progress of neural network-based vocoders, accurate phase reconstruction remains the main challenge limiting both audio quality and modeling efficiency. We introduce PhaseGAN, a lightweight vocoder that addresses this limitation through a "mel $\rightarrow$ Amplitude $\rightarrow$ Phase" reconstruction pipeline. By reconstructing amplitude and phase spectra via distinct methodologies, the proposed PhaseGAN outperforms state-of-the-art baselines while utilizing fewer model parameters and reduced computational requirements. The compact version generates high-fidelity audio with approximately 500K parameters and 1 GMAC computational load, making it highly suitable for real-time applications on edge devices. In addition, our approach exhibits exceptional musical audio synthesis capabilities despite no training on musical data, illustrating unprecedented cross-domain generalization. See <https://github.com/phasegan/phasegan-audio-demo> for demos of our work.

Keywords: neural vocoder · phase reconstruction · speech synthesis · inplace cepstral convolutional recurrent neural network

1 Introduction

Speech and audio synthesis [1,2] have undergone transformative advances with the advent of deep learning, enabling diverse applications from human-computer interaction to creative arts [3,4]. Central to these systems are vocoders—algorithms that synthesize high-fidelity audible waveforms from low-dimensional acoustic representations such as mel-spectrograms.

The performance of a vocoder [5] is evaluated mainly in terms of sound naturalness, intelligibility, resource consumption and synthesis efficiency [6]. Thanks to the rapid development of deep learning, in recent years, neural network-based vocoders (e.g., HiFi-GAN) have far surpassed the early rule-based methods (e.g., Straight [7]) and has dominated this field.

Neural vocoders are typically categorized into five distinct types:

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- Autoregressive models (e.g., WaveNet [8]) generate waveforms sample-by-sample. While capable of synthesizing high-fidelity speech, their prohibitively low computational efficiency renders them impractical for real-time applications.
- Flow-based models (e.g., WaveGlow [9]) utilize invertible transformations to project speech signals into latent spaces. Although enabling parallel waveform synthesis, these models exhibit two key limitations: excessive parameterization and perceptually inferior audio quality compared with autoregressive vocoders.
- GAN [10]-based vocoders (e.g., HiFi-GAN [11]) leverage generator-discriminator adversarial training. They achieve real-time synthesis efficiency and competitive quality, yet face training instability issues and spectral artifacts [12].
- Diffusion models (e.g., DiffWave [13]) reconstruct speech through iterative noise-removal processes. They deliver exceptional stability and perceptual fidelity, but requiring 6-100 sampling iterations.
- Variational autoencoders (e.g., FreeCodec [14]) encode speech signals into disentangled latent representations (e.g., timbre, prosody). These compact models (<5M parameters) support attribute control, though achieve lower perceptual quality compared with GAN and diffusion counterparts.

Compared with other methods, GAN-based vocoders achieve an optimal balance among inference speed [15], synthesis quality, and versatility, which is suitable for real-world applications like voice assistants, accessibility tools, and emotionally expressive AI systems. While challenges such as training instability and artifacts persist, these approaches remain highly practical solutions [16].

The essence of a vocoder is a generative task [8], that is, to generate a sound waveform using low-dimensional acoustic features. From the perspective of the time-frequency domain, the vocoder needs to generate amplitude and phase spectra. If we look at these two parts separately, their generation difficulty is completely different. The reconstruction of the amplitude spectrum is relatively simple, and it is more like an interpolation task, because the low-dimensional acoustic feature input usually contains amplitude information, such as the Mel spectrum. In contrast, the reconstruction of the phase spectrum is more difficult because there is no phase information in the input [17,18,19]. Moreover, the relationship between the amplitude spectrum and the phase spectrum is very complex. One amplitude spectrum corresponds to multiple phase spectra that are exactly the same in auditory perception [20]. Therefore, the reconstruction of the phase spectrum is a generative task. Based on the above thinking, we believe that different methods should be used to reconstruct the amplitude spectrum and the phase spectrum respectively, and this will be helpful for low-resource, efficient and high-fidelity vocoders. We therefore propose a dual-stage vocoder framework: the first stage reconstructs spectral amplitude, while the second generates perceptually coherent phase conditioned on the estimated amplitude spectrum. Compared with the current mainstream T-F domain vocoder methods, we adopted an amplitude-phase decoupling processing approach based on different task difficulties. Moreover, the phase recovery was carried out through gener-

ation, and the entire training process of the model did not involve any phase information of the labels. The main contributions of this work are as follows:

- For the reconstruction of the amplitude spectrum, we first employ a fixed interpolation method to upscale/transform the Mel-spectrum to the linear frequency scale of the amplitude spectrum. We then reconstruct the amplitude spectrum from this interpolated Mel-spectrum using an Inplace Cepstral Convolutional Recurrent Neural Network (ICCRN) model [21]. The ICCRN model is an inplace convolutional architecture, which abandons down and up sampling operations along frequency axis. ICCRN can not only effectively mitigate the impact of artifacts on high-frequency information, but also significantly reduce the number of parameters and calculations.
- To address the issue encountered in phase reconstruction—namely, the "one-to-many" mapping in which a single amplitude spectrum can correspond to multiple perceptually similar phase spectra—we depart from strongly supervised approaches such as Apnet2 [22] and FreeV [23]. Instead, we adopt an adversarial training strategy based on the recently proposed R3-GAN [24], which synthesizes a phase spectrum that closely matches the given amplitude spectrum without relying on any labeled phase information. R3-GAN has demonstrated greater training stability in this setting. To the best of our knowledge, this work represents the first systematic application of such an adversarial training strategy to speech phase reconstruction, thereby establishing a new benchmark for this problem.
- The proposed method significantly outperforms existing approaches in terms of model size, computational efficiency, and synthesized audio quality. Its design enables deployment on resource-constrained embedded devices. Moreover, the method demonstrates strong generalization on both unseen-speaker test sets and cross-domain Chinese singing data, indicating its potential for broader application scenarios. Experiments on end-to-end speech synthesis further confirm the effectiveness of our approach in this task.

2 PhaseGAN

2.1 Overview

As shown in Figure 1, the vocoder’s inference pipeline comprises four steps: First, the Mel-spectrogram is transformed to match the spectral dimension of the original amplitude spectrum via Mel-filter bank pseudo-inversion [30]. Second, an ICCRN reconstructs the amplitude spectrum. Third, the generator synthesizes the phase spectrum conditioned on this amplitude estimate. Finally, we use amplitude and phase spectra to synthesize the audio waveform by inverse Fourier transformation.

2.2 ICCRN

The detailed architecture of ICCRN can be found in our previous work [21]. As an in-place model, ICCRN avoids any frequency downsampling or upsampling operations. This mitigates information loss, prevents the introduction of

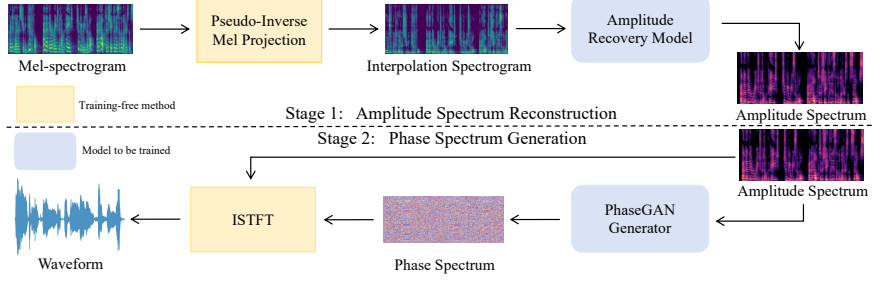


Fig. 1: PhaseGAN Inference Pipeline.

extraneous information, and directly circumvents the artifact problem caused by upsampling operations. Consequently, the frequency dimension of the model remains constant ($f=513$), and the number of output channels is identical across all convolutional layers. To further analyze time-frequency (T-F) domain features, the model applies cepstral analysis. ICCRN also utilizes LSTM [31] along frequency and time axis for dual-path modeling [32]. To ensure the causality of the model, LSTM along time axis is unidirectional, which means no future frame is used. This cepstral analysis capability enables ICCRN to examine the relationship between amplitude and phase information from a harmonic structure perspective; As the core neural network architecture, ICCRN is used in the reconstruction of both the amplitude spectrum and the phase spectrum. Given the distinct mathematical properties and value ranges of amplitude and phase, the corresponding recovery and reconstruction networks differ primarily in the design of their output layers. Further details will be provided in later sections.

2.3 Amplitude Spectrum Reconstruction

Network An interpolation is performed on the 80-dimensional Mel-spectrogram using the pseudo-inverse operation of the Mel-scale filterbank. This process restores the features to their original frequency domain dimensionality ($f = 513$). The interpolated spectrum is subsequently processed by a amplitude recovery model, which is an ICCRN network. Key architectural modifications include: unifying the number of output channels for all convolutional layers to 10 ($c = 10$), and adopting the Softplus activation function for the final output layer. The model takes the interpolation spectrogram as input and generates the optimized amplitude spectrum as output.

Training Loss The mean square error (MSE) loss is employed for amplitude spectrum estimation. Seen in Equation (1):

$$\mathcal{L}_{\text{MSE}} = \frac{1}{T \cdot F} \sum_{t=1}^T \sum_{f=1}^F \left(|\hat{X}(t, f)| - |X(t, f)| \right)^2 \quad (1)$$

where $|X(t, f)|$ and $|\hat{X}(t, f)|$ denote the unit of target and estimate amplitude spectrum at frame t and frequency f , respectively.

2.4 Phase Spectrum Generation

Generator The generator in proposed PhaseGAN similarly employs the ICCRN architecture. The distinction lies in standardizing the number of output channels across all convolutional layers to 20 ($c = 20$). Regarding the output layer design, we draw upon the APNet [33] architecture. The model employs a parallel estimation architecture whose design integrates two linear convolutional layers and the phase formula Φ . This approach is inspired by deriving phase spectra from the real and imaginary components of complex spectra. The pseudo-imaginary component $\hat{I} \in \mathbb{R}^{T \times F}$ and the pseudo-real component $\hat{R} \in \mathbb{R}^{T \times F}$ are the outputs of the two parallel layers. Φ then calculates the phase spectrum $\hat{\theta}$. For $\forall R \in \mathbb{R}$ and $I \in \mathbb{R}$, we define the Φ by Equation (2):

$$\Phi(R, I) = \arctan\left(\frac{I}{R}\right) - \frac{\pi}{2} \cdot \text{Sgn}^*(I) \cdot [\text{Sgn}^*(R) - 1] \quad (2)$$

and $\Phi(0, 0) = 0$. When $x \geq 0$, $\text{Sgn}^*(x)$ equals 1; otherwise, it is -1. Formula Φ strictly restricts the predicted phase to the principal value interval $(-\pi, \pi]$ for direct wrapped phase prediction.

Discriminator The discriminator in PhaseGAN employs a multi-scale architecture (inspired by MelGAN), which takes time-domain speech waveforms as input. It consists of multiple parallel discriminator modules with identical structures, each comprising a three-layer 1D convolutional stack. These modules perform downsampling on input speech features via multi-layer convolutional operations, and their outputs directly drives the adversarial training process for the generator.

Training Loss To train the model of phase generation, we use the R3GAN [24], which significantly enhances training stability by integrating the relativistic paired GAN loss with zero centered gradient penalty mechanism [34]. Experimental results of R3GAN demonstrate that even without the introduction of any specific training techniques, this generative method still exhibits favorable convergence performance and training stability. Due to the high sensitivity of phase information, conventional generative methods often face challenges in achieving stable convergence; therefore, the proposed approach provides a valuable reference for generative phase reconstruction.

The generator's loss function is defined by (3) :

$$\mathcal{L}_G = \alpha \cdot \mathcal{L}_{\text{mrstft}}(\hat{\mathbf{y}}, \mathbf{y}) + \beta \cdot \mathcal{L}_{\text{Adv}}(D(\hat{\mathbf{y}}), D(\mathbf{y})) \quad (3)$$

where G denotes the generator, D represents the discriminator, $\mathbf{y}$ signifies the real audio sample, and $\hat{\mathbf{y}}$ corresponds to the synthesized speech sample. α and β denote the weighting coefficients.

The generator loss consists of adversarial losses and is defined as Equation (4):

$$\mathcal{L}_{Adv}(\mathbf{y}, \hat{\mathbf{y}}) = \mathbb{E}_{(\mathbf{y}, \hat{\mathbf{y}})} \left[\text{softplus}(-(\mathbf{y} - \hat{\mathbf{y}})) \right] \quad (4)$$

where Softplus refers to the softplus activation function, which restricts values to be greater than zero.

To enhance the consistency between phase reconstruction and the recovered amplitude spectrum, a multi-resolution STFT loss is incorporated into the generator's objective function. This loss term indirectly constrains phase gradients, guiding the model to progressively reconstruct phase information that exhibits high correspondence with the amplitude spectrum during training. The mathematical formulation of $\mathcal{L}_{mrstft}$ is defined by (5):

$$\mathcal{L}_{mrstft}(\hat{y}, y) = \frac{1}{K} \sum_{k=1}^K \left[\mathcal{L}_{\text{mag}}^{(k)}(\hat{y}, y) + \mathcal{L}_{sc}^{(k)}(\hat{y}, y) \right] \quad (5)$$

where $K = 3$ indicates the number of distinct resolution configurations. $\mathcal{L}_{\text{mag}}^{(k)}$ is the amplitude spectral loss and $\mathcal{L}_{sc}^{(k)}$ is the spectral convergence loss which defined by (6)–(7):

$$\mathcal{L}_{\text{mag}}^{(k)}(\hat{y}, y) = \left\| |STFT^{(k)}(\hat{y})| - |STFT^{(k)}(y)| \right\|_1 \quad (6)$$

$$\mathcal{L}_{sc}^{(k)}(\hat{y}, y) = \frac{\left\| |STFT^{(k)}(\hat{y})| - |STFT^{(k)}(y)| \right\|_F}{\left\| |STFT^{(k)}(y)| \right\|_F + \epsilon} \quad (7)$$

where $\|\cdot\|_1$ denotes the L1-norm and $\|\cdot\|_F$ denotes the Frobenius norm, which is a matrix norm that quantifies the overall scale of a matrix, with ϵ representing an infinitesimal positive constant to ensure numerical stability.

The discriminator loss is composed of adversarial losses and a zero-centered gradient penalty term. The discriminator's loss function is defined as follows (8):

$$\begin{aligned} \mathcal{L}_D = & \mathbb{E}_{(\mathbf{y}, \hat{\mathbf{y}})} \left[\text{softplus}(- (D(\mathbf{y}) - D(\hat{\mathbf{y}}))) \right] \\ & + \frac{\gamma}{2} \left(\underbrace{\mathbb{E}_{\mathbf{y}} [\|\nabla_{\mathbf{y}} D(\mathbf{y})\|^2]}_{R_1} + \underbrace{\mathbb{E}_{\hat{\mathbf{y}}} [\|\nabla_{\hat{\mathbf{y}}} D(\hat{\mathbf{y}})\|^2]}_{R_2} \right) \end{aligned} \quad (8)$$

where γ denotes the dynamic decay coefficient (characterized by a cosine scheduler) [35], $\nabla_{\mathbf{y}} D(\mathbf{y})$ represents the discriminator's gradient with respect to the real sample input, $\nabla_{\hat{\mathbf{y}}} D(\hat{\mathbf{y}})$ corresponds to the discriminator's gradient with respect to the generated sample input, and $\{R_1, R_2\}$ indicate zero-centered gradient penalties [36].

3 Experimental Setup

3.1 Datasets

To train and evaluate the performance of the proposed PhaseGAN, we employ three datasets.

Single speaker speech synthesis The LJSpeech dataset [37] was used for a single speaker experiment¹. This publicly available benchmark dataset comprises 13,100 audio clips recorded by a single female English speaker, about 24 hours. The audio clips are sampled at 22.05 kHz with 16 bit. We partitioned the dataset into training, validation, and test sets according to Guideline from the open-source VITS repository². This dataset is commonly employed for performance comparison among different neural vocoders.

Unseen speaker speech synthesis VCTK [38], a multi-speaker dataset³, was used to evaluate the generalization ability for unseen speakers outside the training data. It comprises approximately 44,200 short audio utterances from 109 native English speakers with diverse accents, totaling 44 hours. The original audio recordings were sampled at 44 kHz and downsampled to 22.05 kHz in our experiment. To evaluate PhaseGAN’s generalization for Mel-spectrogram inversion on unseen speakers, we randomly selected two speakers (≈ 500 utterances) for testing using models trained exclusively on the single-speaker LJSpeech dataset (without fine-tuning). PhaseGAN’s inversion results were compared against those from HiFi-GAN and FreeV, which were also trained solely on LJSpeech.

Singing Voice speech synthesis To further validate generalization capability, we also evaluate our method on singing voice data that was not included in the training set. We utilized the Opencpop dataset [39]—a Mandarin singing corpus containing 100 pop songs performed by a trained female vocalist⁴. Original audio files were recorded at 44.1 kHz sampling rate. For experimental consistency, we downsampled all audio to 22.05 kHz, removed silent segments using Voice Activity Detection (VAD) [40], and segmented utterances into 10-second clips for model evaluation. Additionally, for fair comparison, we conducted a comparative analysis between the synthesized outputs of our proposed model and those of HiFi-GAN and FreeV. All models were evaluated without any fine-tuning.

3.2 Baseline

To compare vocoder performance, this study chose several representative models for benchmark, including HiFi-GAN [11], iSTFTNet [27], APNet [33], BigVGAN

¹ <https://keithito.com/LJ-Speech-Dataset/>

² <https://github.com/jaywalnut310/vits/tree/main/filelists>

³ <https://datashare.ed.ac.uk/handle/10283/3443>

⁴ <https://wenet.org.cn/opencpop/>

[12], APNet2 [22], Vocos [28] and FreeV [30]. All these models provide officially released pretrained models based on the LJSpeech dataset. However, inconsistencies exist in their official data splits (specifically, the definitions of the training, validation, and test sets). To ensure a fair comparison, all models were uniformly retrained using the open-source data splits provided by the VITS codebase. All baseline models strictly aligned their parameters with the officially released optimal versions.

3.3 Evaluation Metrics

Subjective evaluation To evaluate the audio quality, we crowd-sourced 5-scale MOS tests via Amazon Mechanical Turk. Participants rated speech samples on a scale from 1 ('poor-completely unnatural speech') to 5 ('excellent - completely natural speech'). The MOS scores were recorded with 95% confidence intervals (CI). Raters listened to the test samples randomly, where they were allowed to evaluate each audio sample once. All audio clips were normalized to prevent the influence of audio volume differences on the raters. In the MOS test, 20 listeners participated (for the English test, it was native English speakers; for the singing test, it was 5 native Mandarin speakers), which ensured the reliability. All quality assessments in Section 4 were conducted in this manner, and were not sourced from other papers.

objective evaluation This study employs more objective metrics for thorough evaluation, including the F1 score for voiced/unvoiced classification (V/UV F1, optimal: 1.0), Periodicity error (optimal: 0.0), Pitch RMSE (optimal: 0.0), F0 RMSE (optimal: 0.0), UTMOS (optimal: 5.0) [41], wide-band perceptual evaluation speech quality (WB-PESQ, optimal: 4.5) [42], short time objective intelligibility (STOI, optimal: 1.0) [43], mel-cepstrum distortion (MCD, optimal: 0.0). Specifically, Pitch RMSE and F0 RMSE quantify pitch accuracy, while UTMOS and WB-PESQ target perceptual quality. For spectral fidelity, Periodicity error and MCD evaluate the synthesized speech, whereas STOI and V/UV F1 gauge speech intelligibility.

3.4 Training Setup

We trained all models on the LJSpeech dataset. For our proposed two-stage framework, the Amplitude recovery model (Stage 1) was trained for 1 million steps with 16 minibatch size, while the phase generation model (Stage 2) underwent 1.5 million steps with 16 minibatch size. The baseline model's training steps and minibatch size match those of our proposed method. All models shared the same experimental setup. For feature extraction, a 1024-point Fast Fourier Transform (FFT) was performed, utilizing a Hanning window of length 1024 and a hop size of 256 samples. We used 80 mel-frequency bands with a frequency cutoff of 16 kHz. The sampling rate was 22,050 Hz. Optimization employed AdamW [44] with an initial learning rate of 0.0002, optimizer parameters

(β_1, β_2) set to $(0.8, 0.99)$, and exponential learning rate decay applied with a decay factor of 0.999. Each audio sample was processed via random window cropping, resulting in segments of 16,384 samples (approximately 0.74 seconds). For the multi-resolution STFT loss ($\mathcal{L}_{mrstft}$), this study adopts three distinct time-frequency resolution configurations (window-hop-FFT size): $(400, 80, 512)$, $(800, 200, 1024)$ and $(1600, 400, 2048)$. The weighting parameters α and β in the generator’s loss function are empirically configured at 0.8 and 0.2, respectively.

This study trained two variants of PhaseGAN: standard and lightweight version, named **PhaseGAN** and **PhaseGAN-s**. In the PhaseGAN (standard version), the Amplitude Recovery Module employs a 10-channel ICCRN model, while the Generator employs a 20-channel ICCRN model. In the PhaseGAN-s (lightweight version), the Amplitude Recovery Module and Generator employ 3-channel and 8-channel ICCRN models, respectively. The number of Cepstral Frequency Blocks (CFB) layers were uniformly set to 5 across all models.

4 Experimental Results

4.1 Single-Speaker Expressiveness

Table 1 presents a comparison between the proposed model and baseline methods. Based on the comprehensive comparison in Table 1 of vocoder performance metrics, with only 1.6M parameters, PhaseGAN achieves SOTA in all perceptual quality metrics, like highest UTMOS 4.238, the best subjective human-rated quality, lowest MCD 2.108 indicating superior mel-cepstral fidelity and naturalness. Lowest periodicity 0.088 for PhaseGAN and 0.112 for PhaseGAN-s shows that superior phase estimation reduces "metallic" artifacts common in GAN vocoders. It can also be observed that PhaseGAN obtain lowest Pitch-RMSE 17.926 and F0-RMSE 35.216, which are crucial for natural prosody. PhaseGAN-s offers near-SOTA quality reducing the number of parameters by over 90%, which is suitable for edge devices. At the same time, our subjective MOS (Media Opinion Score) index is also the best. It is noteworthy that the UTMOS score of authentic speech reaches 4.37, demonstrating the high fidelity achieved by our method. We refer to [45] for the reproduction of the baseline model.

4.2 Ablation Study

To validate the effectiveness of key components, we conducted ablation studies on the core loss functions of the training strategy and the essential structure of the ICCRN model. The results are presented in Table 1. Removal of Multi-resolution STFT Loss (w/o $mrstft$): Eliminating this term from the generator’s loss function resulted in significant degradation across all performance metrics. This decline is attributed to reduced fitting accuracy of phase and amplitude information, consequently impairing synthesized speech quality (audible artifacts can be observed in the provided audio samples). Removal of Zero-Centered Gradient Penalty (w/o zgp): Excluding this penalty term from the discriminator’s

Table 1: Objective evaluation results on the LJSpeech single-speaker speech synthesis test set. The last Five rows present results from ablation studies on PhaseGAN, with the best performance highlighted in bold.

Model	Domain	# Mac	# Params	WB-PESQ $\uparrow$	STOI $\uparrow$	MCD $\downarrow$	V/UV	F1 $\uparrow$	Periodicity $\downarrow$	Pitch-RMSE $\downarrow$	F0-RMSE $\downarrow$	UTMOS $\uparrow$	MOS $\uparrow$
HiFiGAN	T	29.44G	13.94M	3.574	0.931	3.641	0.953	0.125	32.279	36.232	4.219	4.233 (+0.03)	
iSTFTNet	T	20.75G	13.26M	3.535	0.932	3.625	0.951	0.124	35.096	37.436	4.236	4.143 (+0.07)	
BigVGAN	T	27.83G	14.01M	3.492(3.519)	0.965	3.168	0.956(0.945)	0.127(0.128)	30.842	37.142	4.137	3.982 (+0.04)	
APNet	T-F	5.99G	72.19M	3.391	0.968	3.285	0.949	0.142	21.252	39.739 (19.83)	3.177	3.783 (+0.04)	
Vocos	T-F	1.11G	13.35M	3.522(3.70)	0.973	2.672	0.957(0.958)	0.115(0.101)	28.185	36.561	3.973(3.734)	4.112 (+0.04)	
APNet2	T-F	2.61G	31.43M	3.492	0.971	2.829(2.078)	0.962	0.108	26.034	40.046(44.33)	3.938	3.874 (+0.05)	
FreeV	T-F	1.51G	18.22M	3.593(3.431)	0.975(0.967)	2.752(3.112)	0.962(0.956)	0.106(0.118)	24.418	39.087(26.40)	4.015	4.017 (+0.06)	
PhaseGAN-s	T-F	1.05G	0.54M	3.672	0.981	2.483	0.952	0.112	20.522	36.623	4.019	3.986 (+0.05)	
PhaseGAN	T-F	6.42G	1.63M	3.926	0.988	2.108	0.969	0.088	17.926	35.216	4.238	4.256 (± 0.05)	
-w/o mrstft.	T-F	6.42G	1.63M	3.482	0.974	2.682	0.956	0.112	24.572	41.072	3.875	3.586 (+0.07)	
-w/o zgp.	T-F	6.42G	1.63M	3.091	0.966	3.062	0.952	0.126	34.221	49.523	3.012	3.014 (+0.04)	
-w/o adv.	T-F	6.42G	1.63M	3.622	0.956	2.345	0.943	0.122	23.426	38.782	3.483	3.743 (+0.05)	
-w/o freq.	T-F	6.32G	1.60M	3.582	0.981	2.463	0.956	0.098	21.516	37.385	3.958	3.686 (+0.04)	
-w/o ceps.	T-F	5.78G	1.60M	3.695	0.983	2.382	0.962	0.102	21.453	39.721	3.534	3.667 (+0.04)	

Note: Values in parentheses are copied from the original papers.

loss function diminished the model’s convergence capability, leading to substantial deterioration in speech synthesis performance metrics. Removal of Adversarial Loss (w/o adv): the accompanying audio samples (refer to provided examples) revealed pronounced mechanical artifacts in the synthesized speech, significantly degrading perceptual quality. This perceptual decline is further corroborated by lower UTMOS scores. Removal of ICCRN Core Modules: The ablation of the freq module (denoting the $\text{LN} \rightarrow \text{Conv}3 \times 1$ module within the CFB) and the ceps module (denoting the Ceps Unit) also resulted in performance degradation. This outcome confirms the critical role both modules play in the task performance of the ICCRN architecture.

4.3 Generalization to Unseen Speakers

To evaluate the generalization capability of the model, this study conducted tests on the VCTK dataset using the model without fine-tuning. As shown in Table 2, the evaluation results demonstrate that the proposed method exhibits robust generalization ability to unseen speakers. The synthesized speech generated by our method demonstrates significantly superior performance across multiple metrics compared to HiFi-GAN and FreeV. This indicates that the phase generation process underlying the method primarily depends on amplitude spectrum information, rather than speaker-specific characteristics. Experimental results indicate that in subjective listening tests, the speech synthesized by our method for unseen speakers outperforms the baseline models in both clarity and naturalness.

4.4 Generalization to Singing Voice

To evaluate the model’s cross-lingual singing voice generalization capability, we conduct zero-shot testing by directly transferring the model trained on the English speech dataset (LJSpeech) to the Mandarin singing voice dataset (Openccpop). As indicated in Table 3, despite the dual disparities in linguistic properties

Table 2: Objective evaluation results on the VCTK unseen-speaker speech synthesis test set (without fine-tuning). The best-performing results are highlighted in bold.

Model	Domain	WB-PESQ $\uparrow$	STOI $\uparrow$	MCD $\downarrow$	V/UV	F1 $\uparrow$	Periodicity $\downarrow$	Pitch-RMSE $\downarrow$	F0-RMSE $\downarrow$	UTMOS $\uparrow$	MOS $\uparrow$
HiFiGAN	T	2.578	0.929	3.212	0.887	0.208	37.109	39.542	3.232	3.886 (± 0.06)	
FreeV	T-F	2.414	0.946	3.802	0.897	0.199	45.382	48.172	3.131	3.764 (± 0.05)	
PhaseGAN-s	T-F	3.041	0.949	2.883	0.936	0.134	18.645	37.452	3.275	3.904 (± 0.07)	
PhaseGAN	T-F	3.199	0.963	2.595	0.941	0.127	16.026	36.772	3.357	4.142 (± 0.07)	

Table 3: Objective evaluation results on the Opencpop singing voice dataset (without fine-tuning). The best-performing results are highlighted in bold.

Model	Domain	WB-PESQ $\uparrow$	STOI $\uparrow$	MCD $\downarrow$	V/UV	F1 $\uparrow$	Periodicity $\downarrow$	Pitch-RMSE $\downarrow$	F0-RMSE $\downarrow$	UTMOS $\uparrow$	MOS $\uparrow$
HiFiGAN	T	2.301	0.821	3.532	0.942	0.268	25.193	39.251	4.012 (± 0.04)		
FreeV	T-F	2.952	0.842	3.011	0.944	0.357	18.312	43.412	3.956 (± 0.07)		
PhaseGAN-s	T-F	3.613	0.892	2.162	0.931	0.257	13.451	41.399	4.132 (± 0.05)		
PhaseGAN	T-F	3.865	0.922	1.512	0.952	0.211	11.961	39.112	4.356 (± 0.07)		

(English vs. Mandarin), articulation modes (speech vs. singing), and acoustic distributions, the proposed method demonstrates exceptional cross-domain robustness. According to the provided audio sample, it can be known that, baseline models exhibit noticeable underperformance on the cross-domain singing dataset, producing rather blurred harmonic structures. In contrast, the proposed method reconstructs results that are comparable in quality to the reference audio, with clear and complete harmonics. For specific auditory examples, please refer to the provided audio samples.

4.5 End-to-End Speech Synthesis

To validate the effectiveness of the proposed model in an end-to-end speech synthesis pipeline, we designed an experimental study. The pipeline consists of two stages: text-to-mel-spectrogram conversion and mel-spectrogram-to-waveform synthesis. First, mel-spectrograms were generated using FastSpeech2, with no structural modifications applied; instead, its widely-used public implementation and pre-trained weights were directly adopted. Subsequently, the generated mel-spectrograms were used as conditional input to the second-stage vocoders, which included the standard and lightweight versions of our proposed model, as well as the baseline systems HiFiGAN and FreeV, for waveform synthesis. The validation data consisted of the test set corresponding to LJSpeech, and all vocoders were used in their pre-trained versions without any fine-tuning, directly synthesizing speech from the mel-spectrograms.

Table 4 presents the results of each model in terms of the UTMOS objective score and subjective mean opinion score (MOS). It can be observed that the proposed method not only demonstrates good generalization ability on unseen speaker and singing datasets, but also shows high compatibility with acoustic model-driven speech synthesis application scenarios. According to the provided audio sample, it can be known that, the proposed method produces virtually

Table 4: The end-to-end speech synthesis quality comparison of different vocoders without any fine-tuning on the LJSpeech test set. The best-performing results are highlighted in bold.

Model	Domain	UTMOS $\uparrow$	MOS $\uparrow$
HiFiGAN	T	4.132	4.145 (± 0.02)
FreeV	T-F	3.952	4.052 (± 0.04)
PhaseGAN-s	T-F	3.978	3.985 (± 0.07)
PhaseGAN	T-F	4.245	4.189 (± 0.06)

no artifacts in the high-frequency range, while reconstructing clearer and more complete harmonics in the low-frequency region compared to the baseline model.

5 Conclusion

This study proposes PhaseGAN, a model capable of synthesizing high-quality, artifact-free speech with ultra-low parameter counts. Experimental results demonstrate its superior synthesis quality over state-of-the-art publicly available models. Inspired by the R3GAN framework from computer vision, we innovatively introduce a phase generation mechanism that effectively addresses the long-standing phase reconstruction challenge in vocoders. To the best of our knowledge, this represents the first successful demonstration of phase-related problem solving via generative adversarial networks. Additionally, PhaseGAN exhibits exceptional generalization capabilities. While maintaining computational complexity comparable to the current optimal public model (FreeV), our lightweight implementation achieves a substantial reduction in parameters while delivering superior audio fidelity. This breakthrough provides new pathways for edge-device natural speech synthesis requiring low latency and minimal memory footprint. And, this work demonstrates a strong potential for industrial scene deployment.

6 Future Work

We will further validate the effectiveness of the proposed method on more diverse datasets, encompassing a wider variety of speakers and languages. In our future work, we will validate our phase reconstruction method on various models to further demonstrate its universality. Additionally, the method of phase reconstruction will be extended to a broader range of application domains, such as codec, speech enhancement, and bandwidth extension.

References

1. Charpentier, F., Moulines, E.: Pitch-synchronous waveform processing techniques for text-to-speech synthesis using diphones. In: First European Conference on

- Speech Communication and Technology, EUROSPEECH 1989, pp. 2013–2019. ISCA (1989)
2. Hunt, A.J., Black, A.W.: Unit selection in a concatenative speech synthesis system using a large speech database. In: 1996 IEEE International Conference on Acoustics, Speech, and Signal Processing, ICASSP '96, pp. 373–376. IEEE Computer Society (1996)
 3. Ren, Y., Ruan, Y., Tan, X., et al.: FastSpeech: Fast, robust and controllable text to speech. *Advances in neural information processing systems* **32** (2019)
 4. Shen, J., Pang, R., Weiss, R.J., et al.: Natural TTS Synthesis by Conditioning Wavenet on MEL Spectrogram Predictions. In: 2018 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2018, pp. 4779–4783. IEEE (2018)
 5. Morise, M., Yokomori, F., Ozawa, K.: WORLD: A Vocoder-Based High-Quality Speech Synthesis System for Real-Time Applications. *IEICE Trans. Inf. Syst.* **99-D**(7), 1877–1884 (2016)
 6. Kalchbrenner, N., Elsen, E., Simonyan, K., et al.: Efficient Neural Audio Synthesis. In: Proceedings of the 35th International Conference on Machine Learning, ICML 2018, pp. 2415–2424. PMLR (2018)
 7. Kawahara, H., Masuda-Katsuse, I., De Cheveigne, A.: Restructuring speech representations using a pitch-adaptive time–frequency smoothing and an instantaneous-frequency-based F0 extraction: Possible role of a repetitive structure in sounds. *Speech communication* **27**(3-4), 187–207 (1999)
 8. van den Oord, A., Dieleman, S., Zen, H., et al.: WaveNet: A Generative Model for Raw Audio. In: Black, A.W. (ed.) *The 9th ISCA Speech Synthesis Workshop*, pp. 125. ISCA (2016)
 9. Prenger, R., Valle, R., Catanzaro, B.: Waveglow: A flow-based generative network for speech synthesis. In: ICASSP IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 3617–3621. IEEE (2019)
 10. Goodfellow, I.J., Pouget-Abadie, J., Mirza, M., et al.: Generative Adversarial Nets. In: *Advances in Neural Information Processing Systems 27*, pp. 2672–2680 (2014)
 11. Kong, J., Kim, J., Bae, J.: Hifi-gan: Generative adversarial networks for efficient and high fidelity speech synthesis. *Advances in neural information processing systems* **33**, 17022–17033 (2020)
 12. Lee, S., Ping, W., Ginsburg, B., Catanzaro, B., Yoon, S.: BigVGAN: A Universal Neural Vocoder with Large-Scale Training. In: *The Eleventh International Conference on Learning Representations, ICLR 2023*. OpenReview.net (2023)
 13. Kong, Z., Ping, W., Huang, J., Zhao, K., Catanzaro, B.: DiffWave: A Versatile Diffusion Model for Audio Synthesis. In: *9th International Conference on Learning Representations, ICLR 2021*. OpenReview.net (2021)
 14. Zheng, Y., Tu, W., Kang, Y., et al.: FreeCodec: A disentangled neural speech codec with fewer tokens. *CoRR* **abs/2412.01053** (2024)
 15. Bak, T., Lee, J., Bae, H., et al.: Avocodo: Generative Adversarial Network for Artifact-Free Vocoder. In: *Thirty-Seventh AAAI Conference on Artificial Intelligence*, pp. 12562–12570. AAAI Press (2023)
 16. Rogachev, A., Ratnikov, F.: Soft Margin Spectral Normalization for GANs. *Comput. Softw. Big Sci.* **8**(1), 12 (2024)
 17. Takamichi, S., Saito, Y., Takamune, N., Kitamura, D., Saruwatari, H.: Phase reconstruction from amplitude spectrograms based on directional-statistics deep neural networks. *Signal Process.* **169**, 107368 (2020)

18. Masuyama, Y., Yatabe, K., Koizumi, Y., Oikawa, Y., Harada, N.: Phase Reconstruction Based On Recurrent Phase Unwrapping With Deep Neural Networks. In: 2020 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP 2020, pp. 826–830. IEEE (2020)
19. Ai, Y., Ling, Z.H.: Low-Latency Neural Speech Phase Prediction Based on Parallel Estimation Architecture and Anti-Wrapping Losses for Speech Generation Tasks. *IEEE ACM Trans. Audio Speech Lang. Process.* **32**, 2283–2296 (2024)
20. Engel, J.H., Hantrakul, L., Gu, C., Roberts, A.: DDSP: Differentiable Digital Signal Processing. In: 8th International Conference on Learning Representations, ICLR 2020. OpenReview.net (2020)
21. Liu, J., Zhang, X.: Iccrn: Inplace cepstral convolutional recurrent neural network for monaural speech enhancement. In: ICASSP IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 1–5. IEEE (2023)
22. Du, H.P., Lu, Y.X., Ai, Y., Ling, Z.H.: APNet2: high-quality and high-efficiency neural vocoder with direct prediction of amplitude and phase spectra. In: National Conference on Man-Machine Speech Communication, pp. 66–80. Springer (2023)
23. Lv, Y., Li, H., Yan, Y., et al.: FreeV: Free Lunch For Vcoders Through Pseudo Inversed Mel Filter. arXiv preprint arXiv:2406.08196 (2024)
24. Huang, N., Gokaslan, A., Kuleshov, V., Tompkin, J.: The GAN is dead; long live the GAN! A Modern GAN Baseline. In: The Thirty-eighth Annual Conference on Neural Information Processing Systems (2024)
25. Kumar, K., Kumar, R., De Boissiere, T., et al.: Melgan: Generative adversarial networks for conditional waveform synthesis. *Advances in neural information processing systems* **32** (2019)
26. van den Oord, A., Li, Y., Babuschkin, I., et al.: Parallel WaveNet: Fast High-Fidelity Speech Synthesis. In: Proceedings of the 35th International Conference on Machine Learning, ICML 2018, pp. 3915–3923. PMLR (2018)
27. Kaneko, T., Tanaka, K., Kameoka, H., Seki, S.: iSTFTNet: Fast and lightweight mel-spectrogram vocoder incorporating inverse short-time Fourier transform. In: ICASSP IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 6207–6211. IEEE (2022)
28. Siuzdak, H.: Vocos: Closing the gap between time-domain and Fourier-based neural vocoders for high-quality audio synthesis. In: The Twelfth International Conference on Learning Representations, ICLR 2024. OpenReview.net (2024)
29. Woo, S., Debnath, S., Hu, R., et al.: ConvNeXt V2: Co-designing and Scaling ConvNets with Masked Autoencoders. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2023, pp. 16133–16142. IEEE (2023)
30. Lv, Y., Li, H., Yan, Y., et al.: FreeV: Free Lunch For Vcoders Through Pseudo Inversed Mel Filter. In: 25th Annual Conference of the International Speech Communication Association, Interspeech 2024. ISCA (2024)
31. DiPietro, R., Hager, G.D.: Deep learning: RNNs and LSTM. In: Handbook of medical image computing and computer assisted intervention, pp. 503–519. Elsevier (2020)
32. Graves, A.: Long short-term memory. *Supervised sequence labelling with recurrent neural networks*, 37–45 (2012)
33. Ai, Y., Ling, Z.H.: APNet: an all-frame-level neural vocoder incorporating direct prediction of amplitude and phase spectra. *IEEE ACM Trans. Audio Speech Lang. Process* (2023)
34. Roth, K., Lucchi, A., Nowozin, S., Hofmann, T.: Stabilizing training of generative adversarial networks through regularization. *Advances in neural information processing systems* **30** (2017)

35. Loshchilov, I., Hutter, F.: SGDR: Stochastic Gradient Descent with Warm Restarts. In: 5th International Conference on Learning Representations, ICLR 2017. OpenReview.net (2017)
36. Roth, K., Lucchi, A., Nowozin, S., Hofmann, T.: Stabilizing Training of Generative Adversarial Networks through Regularization. In: Advances in Neural Information Processing Systems 30, pp. 2018–2028 (2017)
37. Ito, K., Johnson, L.: LJ Speech Dataset. CMU (2017)
38. Veaux, C., Yamagishi, J., MacDonald, K., et al.: CSTR VCTK Corpus: English Multi-speaker Corpus. Centre for Speech Technology Research, University of Edinburgh (2017)
39. Wang, Y., Wang, X., Zhu, P., et al.: Opencpop: A High-Quality Open Source Chinese Popular Song Corpus for Singing Voice Synthesis. In: 23rd Annual Conference of the International Speech Communication Association, Interspeech 2022, pp. 4242–4246. ISCA (2022)
40. Chang, J.H., Kim, N.S., Mitra, S.K.: Voice activity detection based on multiple statistical models. *IEEE Transactions on Signal Processing* **54**(6), 1965–1976 (2006)
41. Saeki, T., Xin, D., Nakata, W., et al.: UTMOS: UTokyo-SaruLab System for VoiceMOS Challenge 2022. In: 23rd Annual Conference of the International Speech Communication Association, Interspeech 2022, pp. 4521–4525. ISCA (2022)
42. Geiser, B., Jax, P., Vary, P., et al.: Bandwidth extension for hierarchical speech and audio coding in ITU-T Rec. G. 729.1. *IEEE Transactions on Audio, Speech, and Language Processing* **15**(8), 2496–2509 (2007)
43. Taal, C.H., Hendriks, R.C., Heusdens, R., Jensen, J.: A short-time objective intelligibility measure for time-frequency weighted noisy speech. In: IEEE international conference on acoustics, speech and signal processing, pp. 4214–4217. IEEE (2010)
44. Loshchilov, I., Hutter, F.: Fixing Weight Decay Regularization in Adam. *CoRR* **abs/1711.05101** (2017)
45. Li, A., Sun, Z., Hao, F., Li, X., Zheng, C.: Neural Vocoders as Speech Enhancers. *arXiv preprint arXiv:2501.13465* (2025)
46. Bak, T., Lee, J., Bae, H., et al.: Avocodo: Generative adversarial network for artifact-free vocoder. In: Proceedings of the AAAI Conference on Artificial Intelligence **37**(11), 12562–12570 (2023)